

1) The name of the defect is crater cracks. The defects are located at the end of the weld where the arc extinguished. When the arc suddenly extinguishes, the weld metal cooled quickly from high temperature. The stresses would not be high if the weld metal could contract freely during the cooling process. However, the thermal contraction could not freely occur due to the constraint imposed by the earlier formed weld. This results in high tensile stresses in the "crater" causing cracking in the location.

2) welding volume rate:

Wire feeding volume rate = Wire cross-sectional area
× feeding speed

$$= \frac{1}{4} \pi \times 3^2 \times 10$$

$$= 22.5 \pi$$

$$\approx 70.69 \text{ mm}^3 \text{ s}^{-1}$$

Wire volume = 50% of the weld volume

∴ welding volume rate:

$$\text{WVR} = 22.5 \pi \times 2$$

$$= 45 \pi$$

$$\approx (41.4 \text{ mm}^3 \text{ s}^{-1})$$

Melting efficiency:

Input heat rate, $HR_w = \text{Welding Rate, } U_m \text{WVR}$

$$HR_w = f_1 f_2 EI = U_m \text{WVR}$$

$$0.6 f_2 (25)(125) = 9.3(45 \pi)$$

$$f_2 = 0.7012034803$$

$$\approx 0.7$$

Travel speed:

Cross-sectional area of weld bead = 30.6 mm^2

$$\text{WVR} = 45 \pi$$

$$VA_w = 45 \pi$$

$$V = \frac{45 \pi}{30.6} = 4.619989196$$

$$\approx 4.62 \text{ mm s}^{-1}$$

2) Feed rate:

Final weld bead is composed of 60% filler and 40% base metal.

$$WVR = 45\pi \text{ mm}^3 \text{ s}^{-1}$$

$$\begin{aligned} \text{Filler wire area, } A &= \frac{\pi D^2}{4} = \frac{\pi (3.0)^2}{4} \\ &= 2.25\pi \text{ mm}^2 \end{aligned}$$

At 60% filler metal, feed rate of filler wire

$$= \frac{WVR \times 60\%}{A} = \frac{45\pi (0.6)}{2.25\pi} = 12 \text{ mm s}^{-1}$$

3) Total heat available for welding:

$$HR = f_1 f_2 EI = U_m A_w v$$

$$\text{Travel velocity, } v = \frac{f_1 f_2 EI}{U_m A_w}$$

$$\begin{aligned} A_w &= 0.35 \times 3.0 \\ &= 1.05 \text{ mm}^2 \end{aligned}$$

$$v = \frac{0.85 \times 0.75 \times (25 \times 10^3) (30 \times 10^{-3})}{5 \times 1.05}$$

$$= \frac{1275}{14}$$

$$\approx 91 \text{ mm s}^{-1}$$

4) Generated heat rate:

$$U_m = K T_m^2$$

$$= 3.33 \times 10^{-6} (1760)^2$$

$$= 10.315008$$

$$\approx 10.32 \text{ J mm}^{-3}$$

$$HR = \frac{U_m A_w V}{f_1 f_2} = \frac{10.32 \times 6 \times 5 \times 7}{0.8 \times 0.6}$$

$$= 4512.816$$

$$\approx 4513 \text{ J s}^{-1}$$

Time to complete surfacing operations:

$$\text{Total length of weld} = 350 \times \frac{2000}{5}$$

$$= 14000 \text{ mm}$$

$$\text{Time to travel} = \frac{14000}{v} = \frac{14000}{7}$$

$$= 2000 \text{ s}$$

$$= \frac{100}{3} \text{ min}$$

$$\approx 33.3 \text{ min}$$